

Impact of milling, enzyme addition, and steam explosion on the solid waste biomethanation of an olive oil production plant

Andres Donoso-Bravo^{1,2} · E. Ortega-Martinez¹ · G. Ruiz-Filippi¹

Received: 16 July 2015 / Accepted: 24 November 2015 / Published online: 15 December 2015
© Springer-Verlag Berlin Heidelberg 2015

Abstract Anaerobic digestion is a consolidated bioprocess which can be further enhanced by incorporating an upstream pretreatment unit. The olive oil production produces a large amount of solid waste which needs to be properly managed and disposed. Three different pretreatment techniques were evaluated in regard to their impact on the anaerobic biodegradability: manual milling of olive pomace (OP), enzyme maceration, direct enzyme addition, and thermal hydrolysis of two-phase olive mill waste. The Gompertz equation was used to obtain parameters for comparison purposes. A substrate/inoculum ratio 0.5 was found to be the best to be used in anaerobic batch test with olive pomace as substrate. Mechanical pretreatment of OP by milling increases the methane production rate while keeping the maximum methane yield. The enzymatic pretreatment showed different results depending on the chosen pretreatment strategies. After the enzymatic maceration pretreatment, a methane production of 274 ml CH₄ g VS_{added}⁻¹ was achieved, which represents an improvement of 32 and 71 % compared to the blank and control, respectively. The direct enzyme addition pretreatment showed no improvement in both the rate and the maximum methane production. Steam explosion showed no improvement on the anaerobic degradability of two-phase olive mill waste; however, thermal hydrolysis with no rapid

depressurization enhanced notoriously both the maximum rate (50 %) and methane yield (70 %).

Keywords Alpeorujo · Anaerobic digestion · BMP · Enzymatic pretreatment

Introduction

Olive oil is a widely used product due to its unique taste and health benefits. Currently, roughly 11 million of hectares of land is devoted for olive tree cultivation worldwide. Around 65 % of the total cultivated area is located in Europe, although America and Oceania have steadily increased its share in the olive oil production field. In Chile, the olive oil production sector has shown a significant and sustained growth in the last decade coming from 3800 to 25,000 ha of cultivated area from 2004 to 2014, respectively. Depending on the type of processing system that process can be classified into two or three phases. Olive Pomace or olive pulp (OP) is the solid waste generated in three-phase system, and alpeorujo or two-phase olive mill waste (TPOMW) is the semi-solid waste generated in a two-phase plant [1].

Anaerobic digestion is a consolidated technology especially in Europe, although increasingly in North and South America as well. Recently, the Chilean government released a 20/25 plan which obligates the electric companies to produce the 20 % of their electricity from non-conventional renewable energy (ERNC) source for 2025. It is in this scenario that the utilization of waste from the agro-industrial sector for energetic conversion, such as biogas, takes an increasing importance. It has been recognized that the anaerobic digestion could be used as a technological option for the stabilization and management

✉ Andres Donoso-Bravo
adonosobravo@gmail.com; andres.donoso@ucv.cl;
andres.donoso@inria.cl

¹ Escuela de Ingeniería Bioquímica, Pontificia Universidad Católica de Valparaíso, Av. Brasil 2085, Valparaíso, Chile

² INRIA-Chile, Communication and Information Research and Innovation Center (CIRIC), Av. Apoquindo 2827, piso 12, Las Condes, Santiago, Chile

of the solid waste coming from olive oil production [2]. Nevertheless, both OP and TPOMW are highly complex due to the high fiber content which can impair the anaerobic biodegradability [3, 4]. Therefore, upstream pretreatment processes may be an interesting alternative for reducing the polymerization grade of the fiber present in the waste and, consequently, increase the biogas production potential and rate since the accessibility of the substrate for the anaerobic microorganism is enhanced.

A wide variety of pretreatment techniques are available with the final goal of improving the anaerobic biodegradability of a specific waste. Mechanical treatment may increase the accessible surface specific area and size of pores as well as reduce the size of the waste that is going to be used as substrate for anaerobic digestion [5]. Thermal/steam explosion pretreatment has been successfully implemented in full-scale wastewater treatment plant to solubilize sewage sludge. This technology has demonstrated its economical and energetic benefit [6]. At lab-scale it has also shown promising results for lignocellulosic biomass [7]. Amid the biological pretreatments, the enzymatic bioreaction is a specific biological reaction which aims to convert specific complex compounds into its monomers. The enzymatic treatment has been evaluated in the degradation of switchgrass by peroxidases and pectinases [8]. In previous studies, the enzyme addition has been done in the same BMP assay, which means that the operating conditions for the enzyme to act are the ones of the BMP test [9]. These conditions do not correspond necessarily to the optimum conditions for the enzyme action.

In the specific case of solid waste from olive oil production, some pretreatment technologies have been proposed and assessed by Ruggeri et al. [10] for the biogas conversion of OP and olive mill wastewater, i.e., for the waste coming from three-phase olive oil production plant. In the case of that study, chemical pretreatment by adding some chemical compounds, ultrasound, and a mix of both were assessed. The aim of this study is the evaluation of the impact of the pretreatment technologies on the methane production using waste from olive oil production, specifically, milling with OP and thermal/steam explosion and enzymatic pretreatment with TPOMW.

Materials and methods

Inoculum and substrate

OP and TPOMW were obtained from three- and two-phase olive oil processing plants, respectively, located in the Region del Maule, south of Chile. The main characteristics and composition of the substrates are given as follows: OP: COD, 835.2 g Kg⁻¹; TS, 452.5 g Kg⁻¹; VS, 445.7 g Kg⁻¹; fat (dry

weight, dw), 8.5 g 100 g⁻¹; ashes (dw), 1.2 g 100 g⁻¹; nitrogen (dw), 0.6 g 100 g⁻¹; fiber (dw), 57.5 g 100 g⁻¹; nitrogen free extract (dw), 32.2 g 100 g⁻¹. TPOMW: TOTAL COD 435.48 g Kg⁻¹, TS 256.67 g Kg⁻¹, VS 247.79 g Kg⁻¹, fats and oil 13.23 g 100 g⁻¹ (dry weight, dw), total nitrogen Kjeldahl 7.95 g 100 g (dw), crude fiber 59.91 g 100 g⁻¹ (dw), and non-nitrogen extract 21.66 g 100 g⁻¹ (dw).

The anaerobic inoculum, (I), (31.38 g TS L⁻¹ and 19.42 g VS L⁻¹) was obtained from a continuous lab-scale stirred tank reactor maintained in mesophilic conditions and feed with sewage sludge at an OLR of 0.8 kg VS m⁻³ day⁻¹. This inoculum was degasified in order to minimize the biomethanization of the remaining organic matter. Secondary sewage sludge for the wastewater treatment plant (WWTP), El Trebal, Santiago de Chile, was used in the thermal pretreatment evaluation as control of the procedure along with TPOMW.

Pretreatments

Milling

Milling of the OP was performed by a manual mill such that the granulometry of the waste is reduced up to 3 mm diameter.

Enzymatic pretreatment

For this study, the commercial Viscozyme[®] L enzyme from Novozymes[®] was chosen. Viscozyme[®] L is a cellulolytic enzyme mixture with several activities (Hemicellulase, cellulase, Xylanase, Pectinase, and others), showing interesting results in Mahdy et al. [11]. The enzyme activity was measured by the filter paper method described by the IUPAC, showing a value of 69.33 ± 0.36 FPU/mL of enzymatic mix. For the enzymatic pretreatment, two different strategies were used in parallel: previous enzymatic maceration and direct enzyme addition into the Biochemical Potential Test (BMP) test.

Enzymatic maceration This pretreatment consisted in a pre-hydrolysis of the TPOMW (0.25 g VS) with an excess of enzyme (5 mL of enzymatic mix) at the enzyme optimum conditions, with 10 mL of acetate buffer (50 mM, pH 4.8), for a period of 24 h of contact time. Enzyme control and reaction blanks were used. Later, the enzyme was inactivated for 40 min at 100 °C in a thermoregulated bath. The effectiveness of this pretreatment was measured as a way of estimating the maximum theoretical benefit of using enzymatic pretreatment at the optimum enzyme conditions.

Direct enzyme addition This pretreatment consisted in the direct addition of the enzyme mix in the BMP test at

different enzyme-to-substrate (E/S) ratios: 3×10^1 , 3×10^0 , 3×10^{-1} , 3×10^{-2} FPU g VS⁻¹ (named EZA, EZB, EZC, EZD, respectively). Enzyme and reaction blanks were also set up, i.e., BMP assay without the addition of substrate and enzyme, respectively.

Thermal and steam explosion pretreatment

The experiment was carried out in an automatic pilot-scale thermal system located at the wastewater treatment plant, El Trebal, Santiago, Chile. This system consisted of a feeding tank, a progressive cavity pump ($P_{\max} = 12$ bar), a steam boiler, a 20-L total volume hydrolysis reactor (working volume = 10 L) connected to a flash tank ($V = 10$ L) with also outlet pipes for steam and hydrolyzed sludge. The pilot plant is equipped with automatic valves that control the steam entrance from the boiler and the sludge exit from the reactor to the flash. A data acquisition and control system is used to measure pressure and temperature and automatically controls the steam inlet and the hydrolyzed sludge exit to the flash. The pump introduces 10 L of sludge into the reactor, and then the steam valve is opened until pressure and temperature reach the set point. At the end of the reaction time, the decompression valve is automatically opened and the hydrolyzed TPOMW flows to the flash tank. The experimental conditions evaluated are shown in Table 1.

Temperature (T) and time (t) determine the severity factor of the treatment. This parameter ($\log R_0$, Eq. 1) was used to express the severity of the pretreatment and is the common term used in steam pretreatments [12].

$$\log R_0 = \log \left(t \times \exp \left(\frac{T - 100}{14.75} \right) \right) \quad (1)$$

Anaerobic biodegradability

Biomethane potential test

The biomethane potential test (BMP) was performed according to [13] guidelines. The assay was mounted in a glass bottle with 100 ml working volume. Sodium bicarbonate was added in a 1 g of NaHCO₃ per gram of volatile solid of substrate. The initial pH of the assay was adjusted to 7.1 ± 0.1 . Blanks were mounted to subtract the biogas produced from the inoculum. The temperature of the assay was 37 °C controlled by a thermoregulated chamber. The methane production was measured by NaOH solution displacement. In the case of the enzymatic maceration pretreatment, Bioprocess Control (AMPTS II) device was used for the BMP assay. All test was prepared in triplicate.

Inoculum activity evaluation

The activity of the inoculums may change over time even though it is taken from the same reactor operated at the same conditions. Inoculum activity controls were also set up in order to know whether the activity of the inoculum used in the different experiments kept the same activity. For this purpose, control of inoculum and soluble starch as substrate, at the same conditions of the BMP test, were set up.

Substrate/inoculum (S/I) ratio assessment and parameter fitting

The S/I optimum estimation was performed with raw and mechanical pretreated OP (rOP and mpOP, hereafter). A certain amount of substrate was added in order to have the

Table 1 Thermal hydrolysis conditions and kinetic parameter values for the Gompertz modified equation from the BMP tests

	T (°C)	t (min)	$\log R_0$	P (ml CH ₄ g VS ⁻¹)	R_m (ml CH ₄ g VS ⁻¹ day ⁻¹)	Lag (days)
Pretreatment control assay						
Raw SS	–	–	–	306.35 ± 7.04	23.00 ± 1.43	9.44 ± 0.40
SE SS	165	30	3.39	355.25 ± 2.59	29.08 ± 0.77	4.82 ± 0.16
Pretreatment test with TPOMW						
Untreated	–	–	–	255.63 ± 2.54	28.19 ± 1.33	7.51 ± 0.22
SE1	141.9	30	2.71	237.10 ± 1.69	24.26 ± 0.76	7.51 ± 0.16
SE2	138.4	15	2.31	228.03 ± 1.54	21.28 ± 0.57	7.33 ± 0.15
SE3	119.5	5	1.27	193.93 ± 1.48	21.26 ± 0.76	8.24 ± 0.17
SE4	166.0	30	3.42	119.41 ± 5.44	4.82 ± 0.41	0.65 ± 0.83
SE5	165.3	15	3.10	248.38 ± 3.11	23.30 ± 1.13	8.63 ± 0.26
SE6	168.8	5	2.72	237.40 ± 2.16	25.42 ± 0.95	9.00 ± 0.18
SE7	171.8	5	2.81	246.01 ± 1.99	22.18 ± 0.68	7.60 ± 0.17
TH	148.1	30	2.89	434.25 ± 5.48	42.73 ± 2.29	6.75 ± 0.28

SE steam explosion, TH thermal (without steam explosion)

following S/I ratios: 0.5–1–3–6.6–9 g VS_S VS_I⁻¹. The inoculum concentration was kept constant at 5 g VS L⁻¹.

For parameter determination, the Modified Gompertz equation (Eq. 2) was used [14], although, in the case of enzymatic pretreatment when the Gompertz equation cannot be applied, a saturation-type equation (Eq. 3) was used:

$$B = P \times \exp\left(-\exp\left(\frac{R_m \times e}{P}(\lambda - t) + 1\right)\right), \quad (2)$$

$$B = P \times \frac{t}{t_m + t}, \quad (3)$$

where B is the biogas produced at time t , P is the maximum biogas production (ml CH₄ g VS⁻¹), R_m maximum biogas production rate (ml CH₄ g VS⁻¹ day⁻¹), λ the lag time (day), and t_m is the time where half of the maximum production is attained. For parameter estimation, a simple least squares criterion, between the simulated and experimental Matlab®, is used for the minimization procedure. The standard deviation was estimated from the covariance matrix of the parameters obtained from the inverse of the Fisher Information Matrix (FIM), which gives a lower bound on the achievable parameter error covariance matrix (CN).

Analytical methods

COD was measured using the Walkley–Black method, and TS and VS by Gravimetric methods; fat, ashes, fiber and nitrogen were estimated in dry basis through methods 96,315; 94,015; 920,169 and 200,111 described in Raposo et al. [15], respectively. Nitrogen free extract was estimated by mass balance.

Results and discussion

Inoculum activity

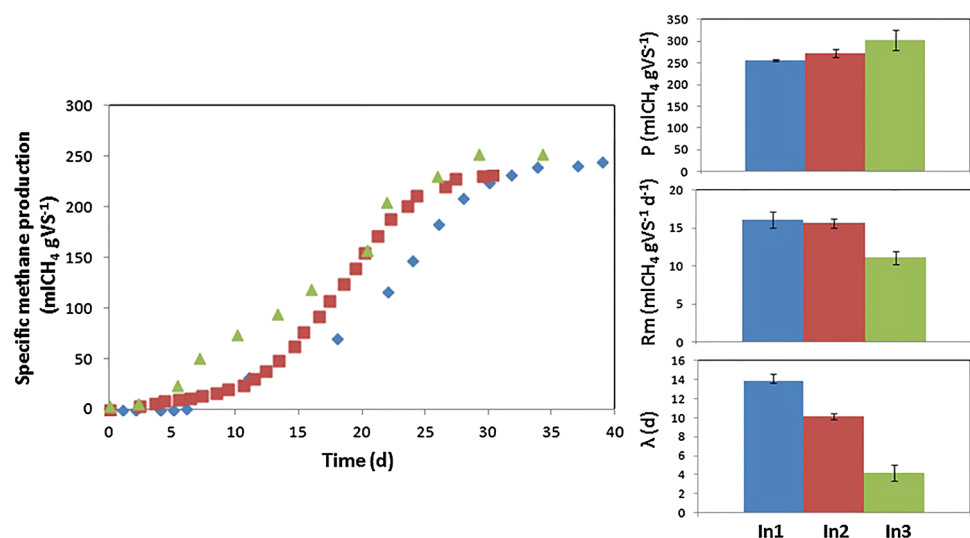
Figure 1 shows the methane production profiles of the inoculum treating soluble starch. It is worth to point out that it corresponds to the anaerobic biomass that was taken from the same lab reactor but at different time (around 3 months apart) since the pretreatment experiences were carried out separately. The methane production profiles are slightly different, and this is demonstrated in the parameter values obtained through the Gompertz equation fitting. The Gompertz modified equation was able to reproduce all the biogas production profiles for every condition, with determination coefficient values always above 0.99. The maximum methane potential increases slightly over time, although not in a significant way. In turn, the maximum production rate dropped during the last pretreatment test. The lag time decreases steadily during the test which shows the inoculum enzymatic machinery get more adapted to hydrolysis macromolecules such as starch.

Mechanical pretreatment of olive pomace (OP)

Substrate/inoculums ratio evaluation

Figure 2 shows the experimental results of the cumulative CH₄ production in the BMP test at different S/I ratios for raw and mechanically pretreated OP, respectively. The Gompertz modified equation was able to reproduce all the biogas production profiles for every condition, with determination coefficient values always above 0.98. The parameters values are shown in Table 2. Overall, it can be

Fig. 1 Accumulated specific methane production profiles of starch degradation and parameters values estimation. Inoculum used for *ln1* (blue): mechanical, *ln2* (red): enzymatic and *ln3* (green): thermal pretreatment (color figure online)



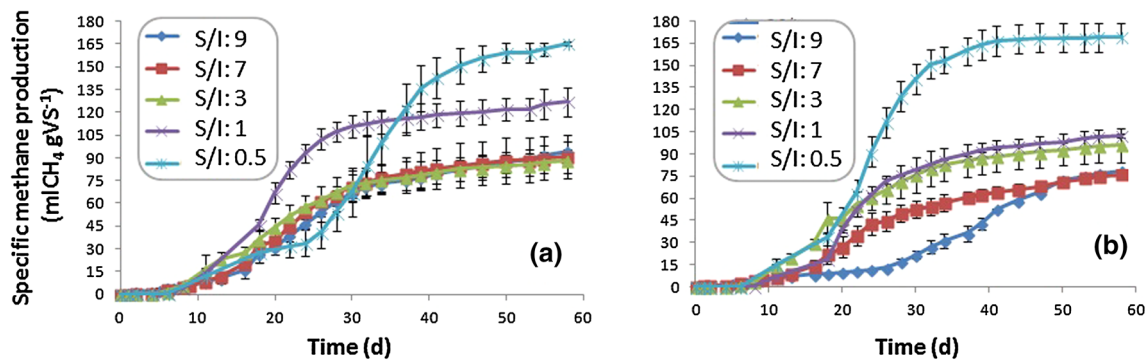


Fig. 2 Accumulated specific methane production profile of **a** raw OP and **b** milled OP at different *S/I* ratios

Table 2 Kinetic parameter values for the Gompertz modified equation at different *S/I* ratios

<i>S/I</i> ratio	<i>P</i> (ml CH ₄ g VS ⁻¹)	<i>R_m</i> (ml CH ₄ g VS ⁻¹ day ⁻¹)	Lag (days)
Raw OP			
0.5	175.85 ± 5.90	6.99 ± 0.50	19.17 ± 0.81
1	122.52 ± 1.03	7.58 ± 0.37	10.91 ± 0.49
3	86.78 ± 0.71	3.62 ± 0.09	7.471 ± 0.32
7	90.32 ± 0.70	3.92 ± 0.09	10.14 ± 0.27
9	92.66 ± 1.27	3.42 ± 0.11	10.25 ± 0.42
Milled OP			
0.5	170.44 ± 1.63	11.25 ± 0.56	15.60 ± 0.44
1	99.20 ± 1.22	5.62 ± 0.33	13.13 ± 0.58
3	94.93 ± 0.85	4.04 ± 0.12	8.14 ± 0.35
7	74.90 ± 1.07	2.71 ± 0.09	9.72 ± 0.45
9	89.50 ± 14.86	2.28 ± 0.07	19.79 ± 0.79

noticed that the *S/I* ratio exerts an important influence on the kinetic of OP degradation which is reflected on the changes of the biogas production profiles. It is clear the best results were obtained for the ratio of 0.5. An extra assay was also performed at ratio of 0.25, but the results were not good enough to be presented. Nevertheless, 0.5 ratio has been identified as a sort of boundary of *S/I* ratio in order to avoid product inhibitions [16].

The greatest methane yield and production rates were achieved at low *S/I*, which is in agreement with those results obtained by [17] who used exhausted sunflower oil cake as substrate. Nevertheless, these results also match those results obtained with other complex substrates such as swine slurry [18]. Nevertheless, it is clear that even for, in theory, a similar substrate like municipal solid waste, the results of the optimum *S/I* may vary depending on the specific composition [19, 20]. These results have been explained by Zhang et al. [21] who state that at low *S/I*, there is a less energy use for cellular division, such that more energy is consumed in other paths such as the one the lead to methane production.

In regard to the parameters values, the *P* values obtained for raw OP and milled OP were much higher to those obtained for [22] who achieved around 80 mL CH₄ g VS⁻¹, although the author did not specify which *S/I* ratio was used. Even though mechanical pretreatment did not show any improvement on the final methane production, as expected, the maximum production rate was significantly improved around 60 % at *S/I* of 0.5 [23]. These results, in practical terms, entail that the retention time necessary to achieve the same methane yield may be reduced by using mechanical pretreatment, and consequently, a smaller full-scale reactor would be needed. Lag time was also reduced after mechanical pretreatment from 20 to 15 days at a *S/I* of 0.5. At greater *S/I*, the lag time seems to drop, which must be related to the amount of substrate added to test the best availability of some readily degradable substrate that can be quickly converted into methane. For the rest of the experiments, a 0.5 substrate/inoculum ratio was used.

Enzymatic pretreatment

Upstream enzymatic maceration

Figure 3 shows the accumulated methane production of the enzymatic pretreatment assay, where $I + E + S^*$ represents the enzymatic macerated substrate degradation, i.e., the assay containing inoculum, inactivated enzyme, and pretreated substrate, $I + S^*$ is the reaction blank containing inoculum and substrate that went through the same pretreatment conditions as $I + E + S^*$ but without the presence of the enzyme, and finally $I + S$ is a BMP test with raw substrate (TPOMW). In Fig. 3a, it is observed that the maceration conditions (optimum conditions for the enzyme) exert an effect on the substrate composition which has to be accounted for. The biogas profile for the $I + S^*$ assay was different in terms of both maximum methane production rate (slope) and the final methane production compared to that for the $I + S$. Therefore, the maceration conditions represent a pretreatment itself since it changes the substrate properties. Figure 3b shows the biogas production profile from the enzyme control ($I + E$) and the enzymatic macerated substrate ($I + E + S^*$). In this case the methane is expressed as ml of methane since there are two sources of organic matter, the substrate and the enzyme which cannot be distinguished, since in the

$I + E + S^*$ test, the kinetic is not necessary the same than the blank due to the co-substrate presence. The simple point-to-point subtraction throughout the test could lead to erroneous values in of the contribution of the enzyme to the methane production. In any case, the results shows that the enzyme is highly degradable (around 70 % is degraded), which is explained by the high volatile solid content (25 % wet basis) composed mostly of sucrose. Therefore, the contribution of the enzyme degradation to the total methane production has to be taken into account due to the fact that most of commercial enzyme contain sugars as a way to stabilize the solution.

The parameters obtained by the equations fitting are presented in Table 3. P is reduced and R_m is increased in the degradation of the substrates that went to maceration conditions. It clearly solubilized some of the TPOMW which is explained by the increase in the methane production rate, but on the other hand, it leads to the formation of some recalcitrant compounds which explained the diminishing in the final methane production. The biogas production profiles of $I + E$ and $I + E + S^*$ present two zones. These zones must be modeled by different models due to the fact that there is a typical biogas profile observed when two substrates are used in a BMP test [24]. The first zone modeled by a saturation equation presents a saturation zone in which the parameters are fairly similar in both

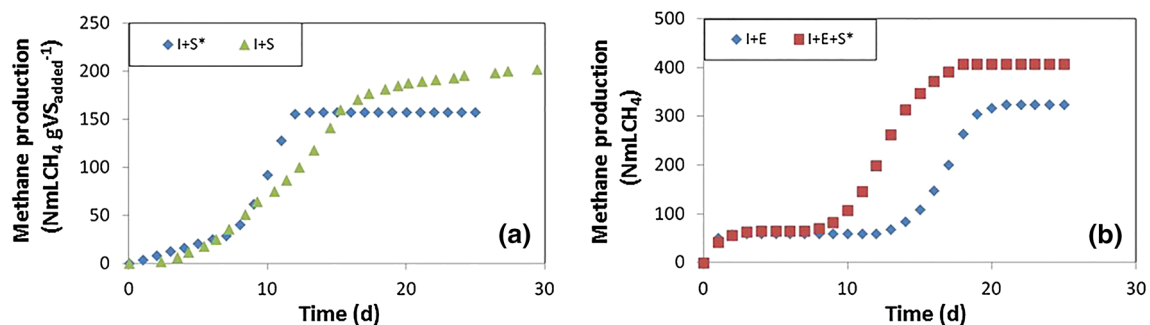


Fig. 3 Evolution of the methane production in the BMP test of **a** test with no enzyme addition ($I + S$) reaction blank ($I + S^*$) and **b** enzymatic macerated substrate ($I + E + S^*$) and the enzyme control ($I + E$)

Table 3 Kinetic parameter values for the Gompertz modified equation at pretreated TPOMW BMP tests

Test	Gompertz modified			Saturation curve	
	P (ml CH ₄ g VS ⁻¹)	R_m (ml CH ₄ g VS ⁻¹ day ⁻¹)	Lag (days)		
$I + S$	207.76 ± 2.90	16.54 ± 0.65	5.49 ± 0.24		
$I + S^*$	160.00 ± 21.50	24.06 ± 3.16	5.89 ± 0.44		
Test	Gompertz modified			Saturation curve	
	P (ml CH ₄)	R_m (ml CH ₄ day ⁻¹)	Lag (days)	P (ml CH ₄)	t_m (days)
$I + E$	272.40 ± 4.76	61.39 ± 4.42	4.42 ± 0.16	62.14 ± 0.11	1.92 ± 0.04
$I + S + E$	351.70 ± 3.04	60.51 ± 1.85	2.64 ± 0.09	73.36 ± 0.38	0.64 ± 0.03

tests, which indicates that in the first day, the part of the enzyme is converted into methane. From that point onwards, a sigmoid behavior was observed, which can be modeled by Gompertz equation. In regard to the effect of the enzymatic maceration, this would lead to a production of 274 ml CH₄ g VS_{added}⁻¹, where around 80 ml of CH₄ was obtained by the biodegradation of TPOMW, which can be estimated by subtracting *P* from *I* + *E* and *I* + *E* + *S** and then dividing for 0.25 g VS of TPOMW added. This represents a significant improvement in comparison with both the raw substrate and the non enzymatic pretreated substrate (*I* + *S**). Overall, the pretreatment improved the final methane production up to 32 and 71 % compared to *I* + *S* and *I* + *S**, respectively. The reduction in the lag phase time is also notorious in the case of *I* + *E* + *S** with almost 50 % of reduction of the parameter value. This entail that, after the first zone (enzyme degradation), the pretreated substrate is rapidly started to be consumed by the anaerobic biomass. Similar results of the enzyme pretreatment of cellulose-rich residues were achieved in batch and continuous reactors by [25].

Modified BMP with enzyme addition

The direct enzyme addition pretreatment showed no improvement in both the rate and the maximum accumulated methane production (Table 4). On the contrary, the more addition of the enzymatic mix the less methane production was attained from the TPOMW, as it seen in the Fig. 4a. The enzyme mix did not produce any improvement of the methane production at these suboptimal conditions of pH and *T*. Furthermore, the maximum methane potential is reduced with each enzyme addition, showing that the enzyme mixture has a negative effect on the TPOMW degradation. It has to be kept in mind that in this pretreatment strategy, the enzyme is not inactivated as the case of the previous one, so the enzyme activity could have harmed the inoculum. The methane production from the blanks test is presented in Fig. 4b, showing that the enzyme mix is anaerobically degradable. The accumulated methane production increases as the enzyme concentration increases, which means that there is no negative effect of the enzyme on the inoculums. The latter demonstrated that catabolite repression takes place, i.e., anaerobic microorganism will

consume the readily substrate first (present in the enzyme mix), provoking some repression the enzymatic activity of the anaerobic microorganism [26]. No methane improvement by adding enzyme into an anaerobic reactor was obtained by Zhang et al. [27] using cattle manure as substrate. Likewise, a negligible improvement was found by Monlau et al. [28] after addition of enzyme in batch and continuous anaerobic reactors treating primary sludge. In other studies where commercial mix were also used, no anaerobic biodegradability improvement has been achieved, which may be explained by the low specificity of those commercial mix as well as by the presence of other degradable compounds [25, 26]. More specific enzymes in regard to the substrate have shown some good results despite working at the BMP test conditions [9].

Thermal pretreatment

The results of the BMP test of thermal pretreated TPOMW are depicted in Fig. 5. As a pretreatment control, a test was carried out with secondary sewage sludge from the WWTP (Fig. 5a). Steam explosion and thermal pretreatment of SS has been widely studied, therefore performing the pretreatment, and the BMP of the raw and pretreated SS will ensure that the pretreatment unit and conditions are working properly. The results of parameters estimation (Table 1), show an improvement of both parameters, *P* and *R_m*, in 16 and 26 %, respectively, and a reduction of the lag phase of 50 %. This indicates that the pretreatment procedure was properly set up. The biogas production from raw TPOMW was higher than the same assay carried out for the enzymatic pretreatment. The changes in the anaerobic biomass activity explained above (Fig. 1) influenced the better results here obtained with the same substrate.

In regard to the TPOMW, neither the thermal pretreatment at different times nor the temperature of the steam explosion yielded an enhancement of the biogas production in comparison with the test with raw TPOMW. Furthermore, the biogas production was significantly impaired at two conditions, SE4 and SE7. In contrast, one condition, TH, with no steam explosion (only cooking) presented the best results with a significant improvement in the maximum biogas rate and total potential. This soft pretreatment condition seems to be the pretreatment condition that suits

Table 4 Kinetic parameter values for the Gompertz modified equation from the direct enzyme addition in the BMP test

	<i>P</i> (ml CH ₄ g VS ⁻¹)	<i>R_m</i> (ml CH ₄ g VS ⁻¹ day ⁻¹)	Lag (day)
Raw TPOMW	207.76 ± 2.90	16.54 ± 0.65	5.49 ± 0.24
EZD	191.47 ± 3.81	13.61 ± 0.68	4.47 ± 0.33
EZC	159.95 ± 2.09	11.04 ± 0.32	5.75 ± 0.20
EZB	118.80 ± 1.77	7.43 ± 0.23	4.29 ± 0.23
EZA	114.64 ± 1.34	11.17 ± 0.47	5.86 ± 0.22

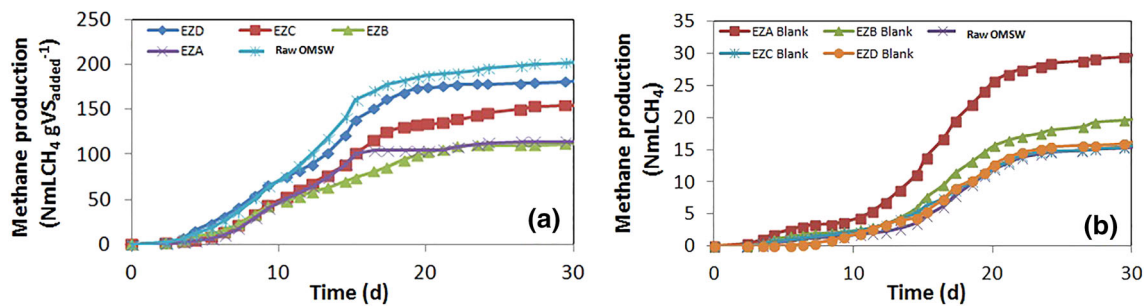


Fig. 4 **a** Evolution of the methane production in the direct enzyme addition pretreatment at different enzyme/substrate (E/S) ratios: raw TPOMW, EZD, EZC, EZB, EZA (0 , 3×10^2 , 3×10^1 , 3×10^0 ,

3×10^{-1} , 3×10^{-2} FPU/g VS, respectively). **b** Methane production of each blank

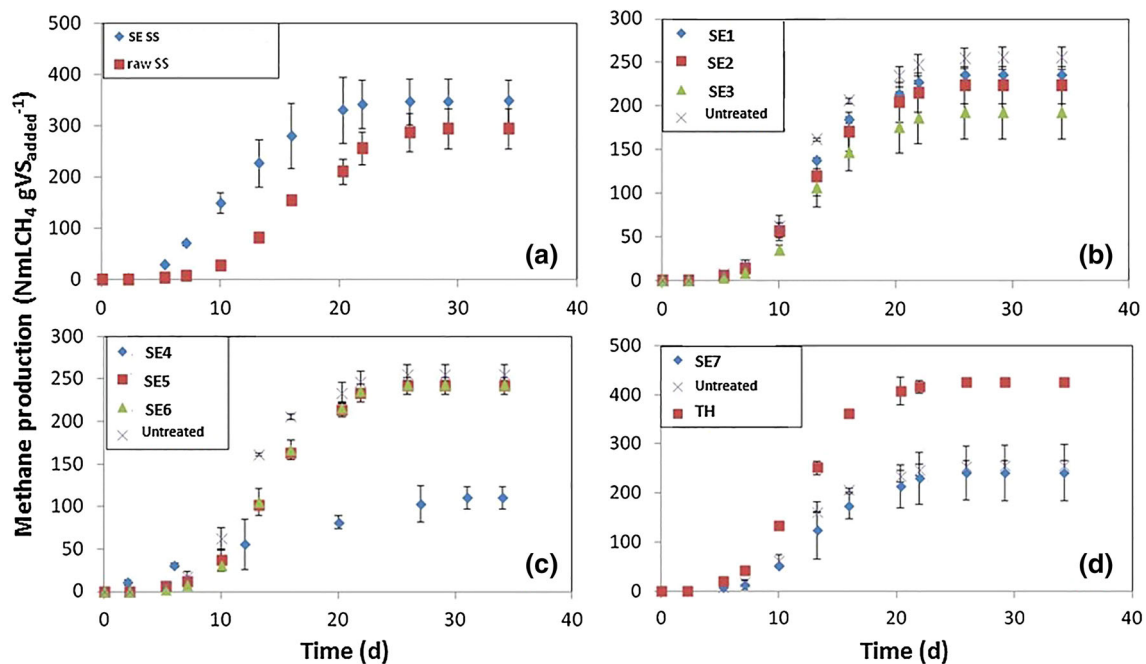


Fig. 5 Evolution of the methane production of: **a** sewage sludge, **b**, **c** and **d** raw and pretreated TPOMW

this substrate the most. Steam explosion may lead to the formation of some inhibitory compounds from the degradation of lignocellulosic biomass, such as furan and phenols, which may actually increase as the explosion temperature rises [29–31]. This was also demonstrated by Pontoni et al. [32] who observed a drop in the methane production of olive oil mill wastewater as temperature increases caused by the presence of acids and phenols.

Conclusions

This study demonstrates that both OP and TPOMW are anaerobically degradable as sole substrate in batch condition. The Gompertz modified equation is able to reproduce

the anaerobic degradation of raw and pretreated OP and TPOMW. *S/I* ratio of 0.5 is the best to be used in anaerobic batch test with OP as substrate. Mechanical pretreatment of OP by milling increases the methane production rate while keeping the maximum methane yield. The enzymatic pretreatment showed different results depending on the chosen pretreatment strategies. The upstream enzymatic maceration lead to an ultimate methane production of $274 \text{ ml CH}_4 \text{ g VS}_{\text{added}}^{-1}$, which represents an improvement of 32 and 71 % compared to the blank and control tests, respectively. The direct enzyme addition pretreatment showed no improvement in both the rate and the maximum methane production. Steam explosion presents no improvement on the anaerobic degradability of TPOMW; however, thermal hydrolysis with no rapid depressurization

enhanced notoriously both the maximum rate (50 %) and methane yield (70 %) in regard to the control.

Acknowledgments This study was funded by Fondecyt Initiation Project No. 11130462 and supported by CIRIC-INRIA-Chile (EP BIONATURE) through Innova Chile Project.

References

- Albuquerque J (2004) Agrochemical characterisation of “alperujo”, a solid by-product of the two-phase centrifugation method for olive oil extraction. *Bioresour Technol* 91:195–200. doi:10.1016/S0960-8524(03)00177-9
- Rincón B, Borja R, Martín MA, Martín A (2009) Evaluation of the methanogenic step of a two-stage anaerobic digestion process of acidified olive mill solid residue from a previous hydrolytic-acidogenic step. *Waste Manag* 29:2566–2573. doi:10.1016/j.wasman.2009.04.009
- Dermeche S, Nadour M, Larroche C, Moulti-Mati F, Michaud P (2013) Olive mill wastes: biochemical characterizations and valorization strategies. *Process Biochem* 48:1532–1552. doi:10.1016/j.procbio.2013.07.010
- Borja R, Rincón B, Raposo F, Alba J, Martín A (2002) A study of anaerobic digestibility of two-phases olive mill solid waste (OMSW) at mesophilic temperature. *Process Biochem* 38:733–742
- Taherzadeh MJ, Karimi K (2008) Pretreatment of lignocellulosic wastes to improve ethanol and biogas production: a review. *Int J Mol Sci* 9:1621–1651. doi:10.3390/ijms9091621
- Pérez-Elvira SI, Fdz-Polanco F (2012) Continuous thermal hydrolysis and anaerobic digestion of sludge. *Energy integration study. Water Sci Technol* 65:1839–1846. doi:10.2166/wst.2012.863
- Horn SJ, Estevez MM, Nielsen HK, Linjordet R, Eijssink V (2011) Biogas production and saccharification of *Salix* pretreated at different steam explosion conditions. *Bioresour Technol* 102:7932–7936. doi:10.1016/j.biortech.2011.06.042
- Frigon J-C, Mehta P, Guiot SR (2012) Impact of mechanical, chemical and enzymatic pre-treatments on the methane yield from the anaerobic digestion of switchgrass. *Biomass Bioenergy* 36:1–11. doi:10.1016/j.biombioe.2011.02.013
- Donoso-Bravo A, Fdz-Polanco M (2013) Anaerobic co-digestion of sewage sludge and grease trap: assessment of enzyme addition. *Process Biochem* 48:936–940. doi:10.1016/j.procbio.2013.04.005
- Ruggeri B, Battista F, Bernardi M, Fino D, Mancini G (2015) The selection of pretreatment options for anaerobic digestion (AD): a case study in olive oil waste production. *Chem Eng J* 259:630–639. doi:10.1016/j.cej.2014.08.035
- Mahdy A, Mendez L, Ballesteros M, González-Fernández C (2014) Enhanced methane production of *Chlorella vulgaris* and *Chlamydomonas reinhardtii* by hydrolytic enzymes addition. *Energy Convers Manag* 85:551–557. doi:10.1016/j.enconman.2014.04.097
- Hendriks ATWM, Zeeman G (2009) Pretreatments to enhance the digestibility of lignocellulosic biomass. *Bioresour Technol* 100:10–18. doi:10.1016/j.biortech.2008.05.027
- Angelidaki I, Alves M, Bolzonella D, Borzacconi L, Campos JL, Guwy AJ, Kalyuzhnyi S, Jenicek P, van Lier JB (2009) Defining the biomethane potential (BMP) of solid organic wastes and energy crops: a proposed protocol for batch assays. *Water Sci Technol* 59:927–934. doi:10.2166/wst.2009.040
- Donoso-Bravo A, Pérez-Elvira SI, Fdz-Polanco F (2010) Application of simplified models for anaerobic biodegradability tests. Evaluation of pre-treatment processes. *Chem Eng J* 160:607–614. doi:10.1016/j.cej.2010.03.082
- Latimer GW (2012) Official methods of analysis of AOAC International, 19th edn. AOAC International
- Raposo F, De la Rubia MA, Fernández-Cegri V, Borja R (2012) Anaerobic digestion of solid organic substrates in batch mode: an overview relating to methane yields and experimental procedures. *Renew Sustain Energy Rev* 16:861–877. doi:10.1016/j.rser.2011.09.008
- Raposo F, Borja R, Martín MA, Martín A, de la Rubia MA, Rincón B (2009) Influence of inoculum–substrate ratio on the anaerobic digestion of sunflower oil cake in batch mode: process stability and kinetic evaluation. *Chem Eng J* 149:70–77. doi:10.1016/j.cej.2008.10.001
- González-Fernández C, García-Encina PA (2009) Impact of substrate to inoculum ratio in anaerobic digestion of swine slurry. *Biomass Bioenergy* 33:1065–1069. doi:10.1016/j.biombioe.2009.03.008
- Boulanger A, Pinet E, Bouix M, Bouchez T, Mansour A (2012) Effect of inoculum to substrate ratio (I/S) on municipal solid waste anaerobic degradation kinetics and potential. *Waste Manag* 32:2258–2265. doi:10.1016/j.wasman.2012.07.024
- Kawai M, Nagao N, Tajima N, Niwa Ch, Matsuyama T, Toda T (2014) The effect of the labile organic fraction in food waste and the substrate/inoculum ratio on anaerobic digestion for a reliable methane yield. *Bioresour Technol* 157:174–180. doi:10.1016/j.biortech.2014.01.018
- Chudoba P, Capdeville B, Chudoba J (1992) Explanation of biological meaning of the $S(o)/X(o)$ ratio in batch cultivation. *Water Sci Technol* 26:743–751
- Tekin AR, Coskun Dalgic A (2000) Biogas production from olive pomace. *Resour Conserv Recycl* 30:301–313
- Zhang Y, Banks CJ (2013) Impact of different particle size distributions on anaerobic digestion of the organic fraction of municipal solid waste. *Waste Manag* 33:297–307. doi:10.1016/j.wasman.2012.09.024
- Fernández-Rodríguez MJ, Rincón B, Fiermoso FG, Jiménez AM, Borja R (2014) Assessment of two-phase olive mill solid waste and microalgae co-digestion to improve methane production and process kinetics. *Bioresour Technol* 157:263–269. doi:10.1016/j.biortech.2014.01.096
- Quiñones TS, Plöchl M, Budde J, Heiermann M (2011) Enhanced methane formation through application of enzymes: results from continuous digestion tests. *Energy Fuels* 25:5378–5386. doi:10.1021/ef2009343
- Philipp B, Schink B (2000) Two distinct pathways for anaerobic degradation of aromatic compounds in the denitrifying bacterium *Thauera aromatica* strain AR-1. *Arch Microbiol* 173:91–96. doi:10.1007/s002039900112
- Sutaryo S, Ward AJ, Møller HB (2014) The effect of mixed-enzyme addition in anaerobic digestion on methane yield of dairy cattle manure. *Environ Technol* 35:2476–2482. doi:10.1080/09593330.2014.911356
- Diak J, Örmeci B, Kennedy KJ (2012) Effect of enzymes on anaerobic digestion of primary sludge and septic tank performance. *Bioprocess Biosyst Eng* 35:1577–1589. doi:10.1007/s00449-012-0748-7
- Zhang Y, Wang L, Chen H (2013) Formation kinetics of potential fermentation inhibitors in a steam explosion process of corn straw. *Appl Biochem Biotechnol* 169:359–367. doi:10.1007/s12010-012-9961-8
- Monlau F, Sambusiti C, Barakat A, Quéméneur M, Trably E, Steyer JP, Carrère H (2014) Do furanic and phenolic compounds of lignocellulosic and algae biomass hydrolyzate inhibit anaerobic mixed cultures? A comprehensive review. *Biotechnol Adv* 32:934–951. doi:10.1016/j.biotechadv.2014.04.007

31. Jacquet N, Quiévy N, Vanderghem C, Janas S, Blecker C, Wathelet B, Devaux J, Paquot M (2011) Influence of steam explosion on the thermal stability of cellulose fibres. *Polym Degrad Stab* 96:1582–1588. doi:[10.1016/j.polymdegradstab.2011.05.021](https://doi.org/10.1016/j.polymdegradstab.2011.05.021)
32. Pontoni L, d'Antonio G, Esposito G, Fabriccino M, Frunzo L, Pirozzi F (2015) Thermal pretreatment of olive mill wastewater for efficient methane production: control of aromatic substances degradation by monitoring cyclohexane carboxylic acid. *Environ Technol* 13–16:1785–1794. doi:[10.1080/09593330.2015.1012179](https://doi.org/10.1080/09593330.2015.1012179)